Example: The Grocery Checkout, a Single-Server Queue

- The scenario we're exploring involves a small grocery store with a single checkout counter.
- Customers arrive at the checkout randomly, with intervals ranging from 1 to 8 minutes apart.
- We're assuming these intervals follow a uniform distribution.

Interarrival Times (minutes)	
Minimum	1
Maximum	8

- Service times at the counter vary from 1 to 6 minutes.

Service Times (Minutes)	Probability	Cumulative Probability
1	0.10	0.10
2	0.20	0.30
3	0.30	0.60
4	0.25	0.85
5	0.10	0.95
6	0.05	1.00

- Our goal is to simulate the arrival and service of 100 customers and analyse various performance measures typical in queueing models.
- However, it's important to note that 100 customers might not provide enough data for reliable conclusions. Depending on our objectives, we might need to increase the sample size or run multiple simulations to improve accuracy.
- Additionally, the starting conditions of the simulation, such as whether the grocery store begins empty or in a steady-state operation, can affect the results. Here, we're simplifying by assuming the store starts empty.

Calculating performance measures

- We are interested in the average waiting time for customers, average time spent in the system, and proportion of time that the server is idle.
- In order to compute these system performance measures, we can use the calculated responses in the table below, which is a snapshot of a few customers. The full simulation table can be found at [DATA_604/Chap2Excel5ed/Example2.5SingleServer.xls at master]

	A	B	C	D	E	F	G	H	I	J
15		**TOTALS**	420		320		163		483	106
16		**AVERAGES**	4.24		3.20		1.63		4.83	1.07
17			**Number of Customers=**		100					
18						**Simulation Table**				
19		**Step**	**Activity**	**Clock**	**Activity**	**Clock**	**Output**	**Clock**	**Output**	**Output**
20			**Interarrival**		**Service**	**Time**	**Waiting Time**	**Time**	**Time Customer**	**Idle Time**
21			**Time**		**Time**	**Service**	**in Queue**	**Service**	**Spends in System**	**of Server**
22		**Customer**	**(Minutes)**	**Arrival Time**	**(Minutes)**	**Begins**	**(Minutes)**	**Ends**	**(Minutes)**	**(Minutes)**
23		1	0	0	2	0	0	2	2	
24		2	5	5	2	5	0	7	2	3
25		3	5	10	4	10	0	14	4	3
26		4	4	14	4	14	0	18	4	0
27		5	2	16	3	18	2	21	5	0
28		6	8	24	2	24	0	26	2	3
29		7	7	31	3	31	0	34	3	5
30		8	8	39	5	39	0	44	5	5
31		9	5	44	1	44	0	45	1	0
32		10	2	46	6	46	0	52	6	1
33		11	1	47	4	52	5	56	9	0

1. Average waiting time (min) = $\frac{total\ time\ customers\ wait\ in\ que\ (minutes)}{total\ number\ of\ customers} = \frac{163}{100} = 1.63$
2. Probability of a customer waiting = $\frac{number\ of\ customers\ who\ wait}{total\ number\ of\ customers} = \frac{46}{100} = 0.46$
3. Probability of an idle server = $\frac{duration\ of\ time\ the\ server\ is\ idle}{total\ duration\ of\ the\ simulation} = \frac{106}{426} = 0.25$
    - This means the busy is idle 25% of the time
4. Average service time = $\frac{total\ number\ of\ service\ minutes}{total\ number\ of\ customers} = \frac{320}{100} = 3.20\ minutes$
    - We can compare this value with the expected average service time by finding the mean of the service-time distribution as follows

$$E(s) = \sum_{0}^{\infty} sp(s)$$

- Where s is a service time and p is the probability of that service time. E.g. service time of 6 minutes has a probability of 0.05.
- Thus Expected service time = 1(0.10) + 2(0.20) + 3(0.30) + 4(0.25) + 5(0.10) + 6(0.05) = 3.2 minutes.
- Although in this simulation run the expected service time is exactly the same as the average service time, this is generally unlikely (Here it is pure coincidence 😄)

5. Average interarrival time = $\frac{sum\ of\ all\ interarrival\ times}{total\ number\ of\ arrivals - 1}$

$$= \frac{420}{99} = 4.24\ minutes$$

- Since the first arrival occurs at time 0, we subtract 1 from the denominator in the equation above (there is no interarrival time before the first customer), giving us 99 instead of 100.
- We can compare the result to the expected average interarrival time by finding the mean of the discrete uniform distribution of interarrival times whose endpoints are a = 1 and b = 8

$$E(A) = \frac{a+b}{2}$$

$= \frac{1+8}{2}$ = 4.5 minutes

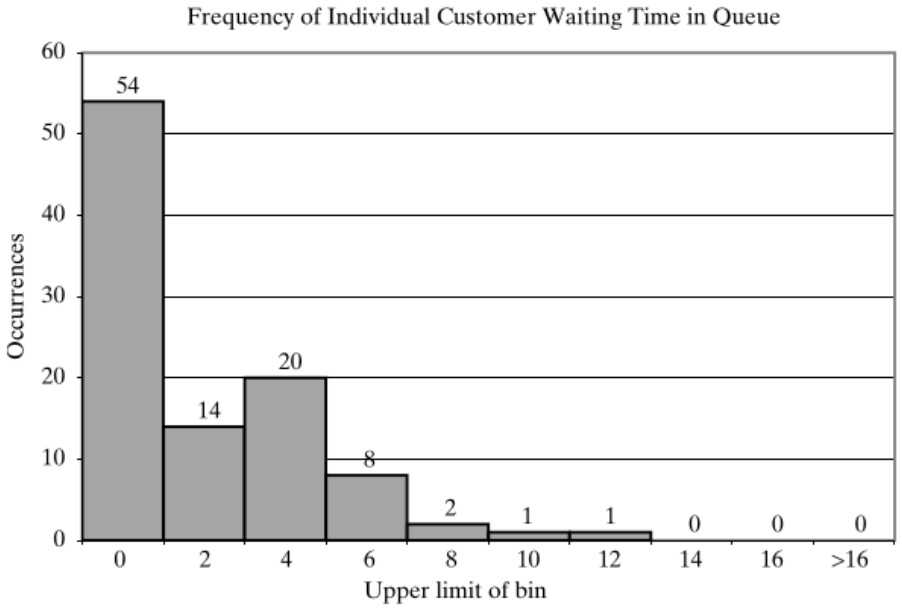
6. Average time of customers who wait = $\frac{sum\ of\ waiting\ times}{total\ number\ of\ waiting\ customers} = \frac{163}{46} = 3.54\ minutes$
   - Note that this is not the same as average waiting time (see performance measure (1))
7. Average time spent in the system = $\frac{total\ time\ customers\ spend\ in\ the\ system}{total\ number\ of\ customers} = \frac{483}{100} = 4.83\ minutes$
   - We can also compute this as average waiting time + average service time (see (1) and (4))

Experimenting with Variability in Queueing Systems (Tutorial).

- Now, we're curious to investigate how the average time customers spend waiting in the queue changes from day to day.
- To do this, we'll run an experiment using our simulation worksheet.
- Each time we run the 'One Trial' worksheet with 100 customers, it represents one day of operation.
- Running this worksheet 50 times simulates 50 days of operation and corresponds to using the 'Experiment' worksheet with the "Number of Trials" set to 50.

Analysing Results from One Trial:

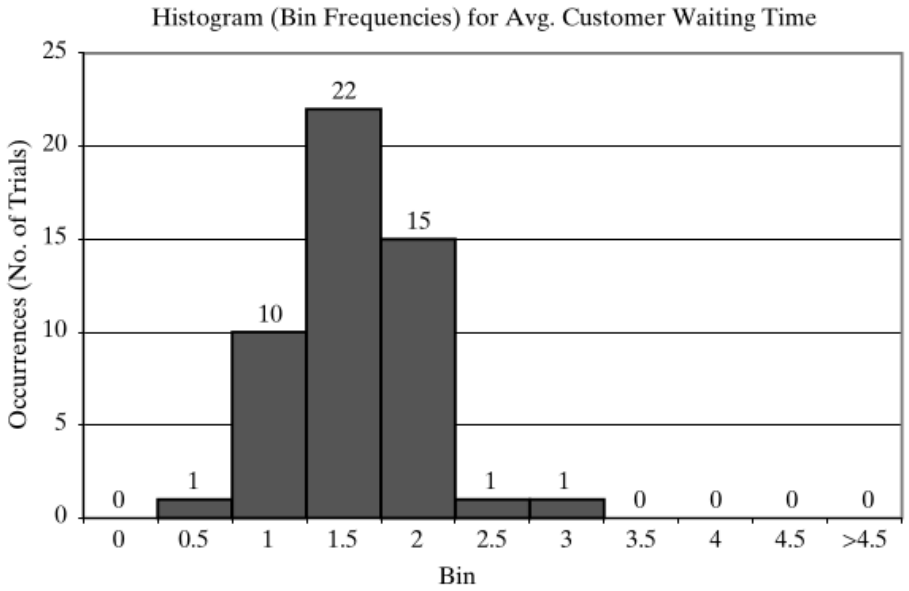
- Let's take a look at the frequency of waiting times for the first trial of 100 customers, shown in figure below:



- In the histogram, we can see that 54% of customers didn't have to wait, and among those who did, 34% waited less than four minutes.
- Remember, these results represent just one day, and we need more data to draw meaningful conclusions.

Running Multiple Trials:

- To get a clearer picture, we'll switch to the 'Experiment' sheet and set the "Number of Trials" to 50.
- Clicking the 'Reset Seed & Run' button generates 50 estimates of the average time customers spend waiting in the queue, each averaged over 100 customers.
- The overall average waiting time across 50 trials turns out to be 1.32 minutes.
- Figure below displays a histogram showing the distribution of average waiting times over the 50 trials. This gives us insights into how the waiting time varies from day to day.



- Try and experiment with different numbers of trials. What if we run 400 trials or just 10 trials instead of 50?
- How does changing the number of trials affect the distribution shown in Figure 10?

References

Banks, Carson II, J., Nelson, J., Nicol, B. (2014). Simulation examples in spreadsheet. In *Discrete-Event System Simulation* (pp. 45-51). Pearson.